

## MODULE 1

# Understanding Defense

Network Defense | Cisco Networking Academy

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## 1.1 Defense-in-Depth

### Topics Covered:

- 1.0 Introduction
- 1.1 Defense-in-Depth
  - 1.1.1 Assets, Vulnerabilities & Threats
  - 1.1.2 Identify Assets & Avatar
  - 1.1.4 Asset Classification & Standardization
  - 1.1.6 Asset Lifecycle Stages
  - 1.1.8 Identify Vulnerabilities & Threats
  - 1.1.10 Security Onion & Security Artichoke
  - 1.1.11 Defense-in-Depth Strategies

# 1.0 Introduction

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## What is Network Defense?

- Protecting networks from unauthorized access, misuse, or denial of service
- Combines hardware, software, and security policies
- Both proactive (prevent) and reactive (respond) measures
- An ongoing process not a one-time activity
- Requires understanding how attackers think and operate

## Learning Objectives

- Define assets, vulnerabilities, and threats
- Explain the Defense-in-Depth strategy
- Classify and manage organizational assets
- Identify and assess vulnerabilities and threats
- Apply layered security strategies
- Understand the Security Onion and Artichoke models

# 1.1 Defense-in-Depth

*Multiple independent layers of security protecting the same asset*

## What it is

- A security strategy that stacks multiple independent controls
- If one layer fails, the next one stops the attacker
- No single control is sufficient on its own
- Each layer uses different tools and techniques
- Slows attackers down gives time to detect and respond
- Limits damage even when one control is bypassed

## The Seven Layers

- Policies & Procedures
- Physical Security
- Perimeter Security
- Network Security
- Host Security
- Application Security
- Data (the core asset being protected)

# 1.1.1 Assets, Vulnerabilities & Threats

*The three fundamental concepts of security risk*

## Assets

- Hardware: servers, routers, switches
- Software and applications
- Data and databases
- People and intellectual property
- Organizational reputation

## Vulnerabilities

- Unpatched or outdated software
- Misconfigured devices or services
- Weak or default passwords
- Unnecessarily open ports
- Missing encryption

## Threats

- Malware: viruses, ransomware, worms
- Phishing and social engineering
- Insider threats
- DoS and DDoS attacks
- Nation-state attackers

# 1.1.2 & 1.1.3 Identify Assets & Avatar

*Asset discovery and user profiling*

## 1.1.2 Identify Assets

- Build a complete inventory of all organizational assets
- Use automated discovery tools such as Nmap
- Categorize assets: hardware, software, data, personnel
- Document ownership, location and business value
- Assign impact ratings based on criticality
- Review and update the inventory regularly

## 1.1.3 Avatar (User Profiling)

- Create profiles of users who access organizational systems
- Capture role, access level and typical usage patterns
- Establish a baseline of normal user behavior
- Detect anomalies that may indicate insider threats
- Supports Least Privilege and Zero Trust principles
- Feeds into SIEM and behavioral analytics platforms

# 1.1.4 & 1.1.5 Asset Classification & Standardization

Organizing assets and enforcing consistent security baselines

## 1.1.4 Asset Classification

**Top Secret**

Maximum protection required

**Confidential**

Restricted to authorized personnel

**Internal**

Not for public release

**Public**

Freely shareable

*Classification governs access, storage, transmission and disposal of each asset.*

## 1.1.5 Asset Standardization

- Enforce consistent hardware and software specifications
- Simplifies patch and configuration management
- Makes deviations from the baseline easy to detect
- Aligns with benchmarks: CIS, NIST, ISO 27001
- Use golden images for rapid and secure deployments
- Reduces attack surface through uniformity
- Speeds up incident response and forensics



## 1.1.6 Asset Lifecycle Stages

*Managing every asset from procurement through to secure disposal*

### 1. Procurement

- Evaluate vendors against security criteria
- Include security clauses in contracts
- Verify integrity on delivery

### 2. Deployment

- Apply security baseline before going live
- Harden the system configuration
- Register in the asset inventory

### 3. Operation

- Monitor continuously and patch
- Audit access logs and usage
- Reassess configuration regularly

### 4. Maintenance

- Repair and upgrade as needed
- Re-evaluate security posture
- Keep documentation current

### 5. Disposal

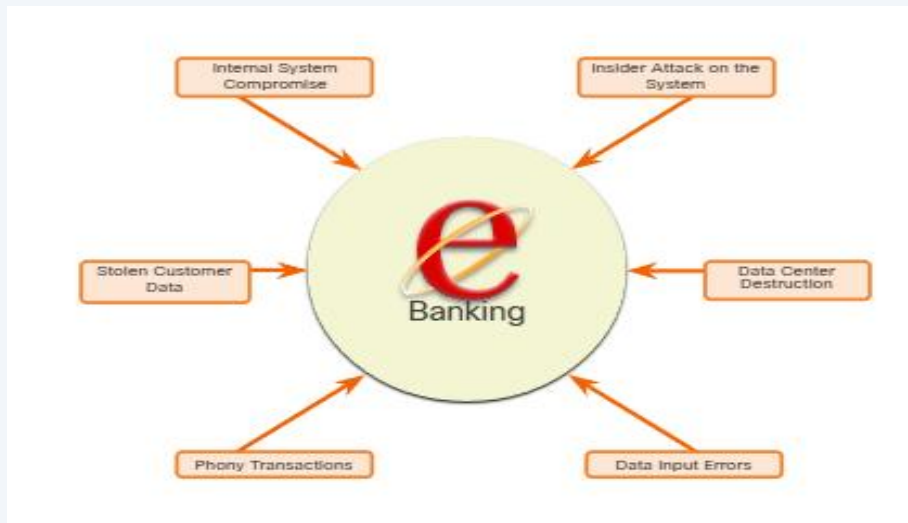
- Securely wipe or destroy media
- Deregister from inventory
- Comply with disposal regulations

# 1.1.8 & 1.1.9 Identify Vulnerabilities & Threats

*Recognizing weaknesses and the dangers that can exploit them*

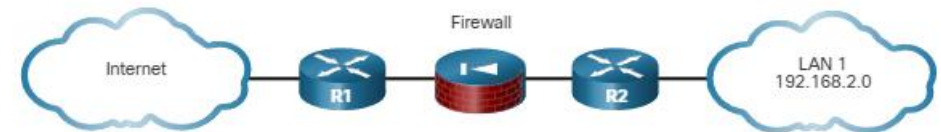
## 1.1.8 Vulnerabilities

- Technical: unpatched software, open ports, weak configs
- Operational: poor procedures, missing audits, no training
- Human: negligence, susceptibility to social engineering
- Scored using CVSS (Common Vulnerability Scoring System)
- Managed and tracked in a formal risk register
- Tools: Nessus, OpenVAS, Qualys, Nmap



## 1.1.9 Threats

- External: hackers, nation-states, organized crime
- Internal: disgruntled employees, accidental data leaks
- Malware: viruses, ransomware, trojans, rootkits
- Network-based: MitM, DoS/DDoS, packet sniffing
- Social engineering: phishing, vishing, pretexting



A router first screens the traffic before forwarding it to a dedicated firewall appliance, for example, the Cisco ASA.

$$\text{Risk} = \text{Threat} \times \text{Vulnerability} \times \text{Asset Value}$$

# 1.1.10 Security Onion & Security Artichoke

*Two models for thinking about layered defense*

## Security Onion Traditional Networks

- Outer-to-inner layers protect the core data asset
- Attacker must defeat every layer to reach the centre
- Each layer detects and blocks before passing deeper
- Well-suited for traditional on-premises environments
- Layers: Policies, Physical, Perimeter, Network, Host, Application, Data

## Security Artichoke Modern / Cloud

- Designed for cloud, mobile and distributed networks
- "Leaves" represent multiple overlapping attack surfaces
- No single clear perimeter unlike the traditional Onion
- Attackers can bypass outer leaves to reach the core
- Security must exist at every point, not just the edge
- Underpins Zero Trust Architecture (ZTA)
- Better suited for today's hybrid environments

## 1.1.11 Defense-in-Depth Strategies

*Six principles for implementing layered security effectively*

### Layering

Stack multiple independent controls. A single failure must not compromise the whole system.

### Limiting

Apply Least Privilege grant users access only to what they need for their role.

### Diversity

Use different vendors and control types so one weakness cannot defeat all layers.

### Obscurity

Conceal system details using port hiding, honeypots and deception technologies.

### Simplicity

Keep designs simple complex systems introduce more gaps and are harder to audit.

### Education

Train all users, enforce clear security policies and run regular awareness programs.

# Key Takeaways

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Assets must be identified, classified, valued and managed across their full lifecycle.



Defense-in-Depth layers controls so a single failure cannot expose the core asset.



$\text{Risk} = \text{Threat} \times \text{Vulnerability} \times \text{Asset Value}$  all three must be continuously assessed.



Security Onion suits traditional perimeters; the Artichoke suits modern cloud environments.



Six core strategies: Layer, Limit, Diversify, Obscure, Simplify, Educate.